TABLE 5
ENCODER VALIDATION PERFORMANCE AT −30 DB.

Background	P_d (%)	False alarms N_{FP}/N_0	Bearing MAE (°)	Precision	Recall	F1
White noise	100.00	0/96	1.26	100.00	93.64	96.72
Mixed background	98.96	0/96	0.22	81.75	81.83	81.79

Each row uses a separately trained encoder and 96 target-present plus 96 target-absent validation observations. Bearing MAE includes all target-present observations. Frequency Precision, Recall, and F1 follow Section 6.2 and are percentages.

TABLE 6
SYSTEM TEST PERFORMANCE UNDER THE MIXED BACKGROUND.

SNR (dB)	P_d (%)	P_{fa} (%)	Bearing MAE (°)	Precision (%)	Recall (%)	F1 (%)	SNR MAE (dB)	Range MAE (km)
0	100.00	–	0.000	77.54	35.74	48.93	0.000	8.410
−10	100.00	–	0.000	65.74	38.25	48.36	1.979	8.222
−20	100.00	–	0.260	65.76	45.34	53.68	0.451	9.174
−25	100.00	–	0.260	64.81	51.84	57.61	0.625	9.462
−30	100.00	–	0.521	63.20	56.78	59.82	0.105	8.749
Overall	100.00	0.00	0.208	67.11	43.95	53.11	0.632	8.803

Frequency Precision, Recall, and F1 follow Section 6.2 and are percentages. Each SNR row contains 288 target-present observations. False alarms are evaluated once on 288 shared target-absent observations, giving 0/288, and are shown only in the overall row. SNR and range have numerical outputs for 287/288 observations at −30 dB and 288/288 at every other SNR, giving 1,439/1,440 overall.

remove absolute received level as a unique range cue; any useful range information must remain in relative spatial or propagation structure.

6.4.2 Structured-Report Examples

The four scalar fields satisfy the joint type and null-value rules in 1,727 of 1,728 reports. One target-present observation at −30 dB has null SNR and range, while its detection, bearing, and frequency outputs remain available.

In Fig. 5, each displayed pair contains a value and its support score. Scalar scores are geometric means of the conditional probabilities of the tokens forming the reported field value. Frequency scores are the sigmoid outputs at the corresponding bins. Scores are truncated to two decimal places for display.

The positive example retains all 23 frequency candidates. Six of the ten highest-scoring candidates lie within ±2 Hz of integer multiples of a possible 12.2 Hz fundamental, corresponding to a repetition period of approximately 82 ms. The illustrative explanation relates this pattern to a possible periodic source component and lists the other four candidates separately. The predicted bearing of 60° agrees with the reference, whereas the predicted range of 10 km differs from the reference range of 25 km. The report lists the predicted attributes and the proposed spectral relation for inspection.

7 CONCLUSION

In this paper, we have developed a physics-guided audio-language approach for underwater passive target detection, building on the unified framework introduced in Part I. The proposed acoustic encoder combines spatial, temporal, and spectral information from multichannel observations, using directional features to guide the extraction of weak acoustic structures. These representations are aligned with a pretrained language model and supplemented with environmental information to produce detection descriptions containing target-presence decisions and associated physical attributes. A physics-parameterized simulation pipeline provides multichannel recordings with traceable supervision, supporting both acoustic representation learning and language adaptation. Experiments with simulated targets under white and mixed backgrounds provide initial evidence for the feasibility of the proposed approach to passive acoustic detection. Together with the recognition study in Part I, this work extends the framework toward a more complete account of underwater acoustic targets, from detecting their presence to describing their characteristics.

Future work may examine the approach under changing propagation conditions, diverse sensor configurations, and continuous observations of moving targets, including validation with field recordings. It will also investigate how generated descriptions relate to the acoustic information underlying detection decisions.

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